

SEPHORA

SQL ANALYSIS & FINDINGS REPORT

Product & Customer Sentiment Analytics

Seema Kumari | Data Analyst

PostgreSQL • Python • Power BI • Excel

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Intro

I ran this analysis in PostgreSQL against two tables I cleaned and loaded myself — products (8,216 rows) and reviews (601,131 rows). Below are the actual queries I wrote, the results, and what I took away from each one. Every finding here is backed by a query, not a guess.

Q1 — Which categories have the highest vs lowest average rating?

```
select primary_category,
round(avg(rating)::numeric,2) as avg_rating,
count(*) as product_count
from products
group by primary_category
order by avg_rating desc;
```

Category	Avg Rating	Product Count
Gifts	4.56	4
Men	4.50	59
Tools & Brushes	4.27	50
Skincare	4.23	2351
Fragrance	4.23	1339
Hair	4.20	1416
Bath & Body	4.20	395
Makeup	4.15	2328
Mini Size	4.01	274

Finding: Gifts and Men top the list but have tiny sample sizes (4 and 59 products) — not something to build strategy on. Among the big categories, Makeup is the lowest rated (4.15) despite being the second-largest category by product count (2,328 products) — that's a real signal, not noise, since the sample is huge.

Q2 — Does higher price actually correlate with higher rating?

```
select
  case
    when price_usd < 20 then 'under $20'
    when price_usd between 20 and 49.99 then '$20-$49'
    when price_usd between 50 and 99.99 then '$50-$99'
    else '$100+'
  end as price_band,
  round(avg(rating)::numeric,2) as avg_rating,
  count(*) as product_count
from products
group by price_band
```

```
order by min(price_usd);
```

Price Band	Avg Rating	Product Count
Under \$20	4.02	1169
\$20-\$49	4.20	4597
\$50-\$99	4.26	1571
\$100+	4.27	879

Finding: Rating climbs steadily with price — 4.02 under \$20, up to 4.27 at \$100+. So price and rating are genuinely linked, which makes the next query more important: which specific expensive products break this pattern? Those are the real risk flags, not just “expensive = risky.”

Q3 — Risk products: high price, low rating, real review volume

```
select product_name, brand_name, primary_category, price_usd, rating, reviews,
       round((price_usd * reviews)::numeric, 0) as revenue_exposure
from products
where price_usd > 50 and rating < 3.8 and reviews >= 10
order by revenue_exposure desc
limit 10;
```

Product	Brand	Category	Price	Rating	Reviews	Exposure
Love, Don't Be Shy	KILIAN Paris	Fragrance	275	3.67	1256	345,400
C-Tango Vitamin C Eye Cream	Drunk Elephant	Skincare	64	3.75	1484	94,976
REPLICA Autumn Vibes	Maison Margiela	Fragrance	160	3.74	569	91,040
Neroli Portofino	TOM FORD	Fragrance	295	3.47	277	81,715
MÉTALLIQUE	TOM FORD	Fragrance	235	3.65	225	52,875
Trinity + Eye and Lip Enhancer Bundle	NuFACE	Skincare	449	3.44	105	47,145
Trinity Facial Toning + Wrinkle Reducer	NuFACE	Skincare	429	3.44	105	45,045

Product	Brand	Category	Price	Rating	Reviews	Exposure
Needles No More Wrinkle Smoothing Cream	Dr. Brandt Skincare	Skincare	89	3.52	506	45,034
Touch Home Permanent Hair Removal Device	iluminage	Skincare	449	3.67	100	44,900
Missing Person Eau de Parfum	PHLUR	Fragrance	96	3.43	433	41,568

Finding: I added a 'revenue exposure' column (price × review count) to rank risk by actual business impact, not just rating. Two clear clusters show up: high-end fragrance (KILIAN, TOM FORD, Maison Margiela, PHLUR) and skincare devices (NuFACE, iluminage). KILIAN's 'Love, Don't Be Shy' tops the list at ~\$345K exposure — a \$275 fragrance with over 1,200 reviews averaging 3.67. That's not a one-off complaint, that's a pattern at scale. Recommend a reformulation/quality review focused on these two segments specifically, not the whole catalog.

Q4 — Worst-performing brands (min 15 products, to avoid small-sample noise)

```
select brand_name,
       round(avg(rating)::numeric, 2) as avg_rating,
       count(*) as product_count,
       round(avg(price_usd)::numeric, 2) as avg_price
from products
group by brand_name
having count(*) >= 15
order by avg_rating asc
limit 15;
```

Brand	Avg Rating	Products	Avg Price
FORVR Mood	3.60	23	43.17
Slip	3.63	15	56.40
INC.redible	3.69	15	12.93
Floral Street	3.70	19	49.47
PHLUR	3.77	19	60.68
Isle of Paradise	3.80	19	26.95
NAILS INC.	3.82	25	17.06
TAN-LUXE	3.86	15	46.13
KAYALI	3.86	28	66.82

Brand	Avg Rating	Products	Avg Price
Supergoop!	3.87	39	32.44
Glossier	3.89	33	23.55
Jo Malone London	3.90	80	100.84
Maison Louis Marie	3.90	30	53.30
Juliette Has a Gun	3.91	32	80.53
goop	3.91	31	62.06

Finding: FORVR Mood is the single worst-rated brand with real volume (3.60 across 23 products). More importantly — fragrance brands dominate this bottom-15 list (PHLUR, Floral Street, Jo Malone, Juliette Has a Gun, KAYALI, Maison Louis Marie, goop). That's the third query in a row pointing at fragrance as a category-wide weak spot, not a one-brand problem.

Q5 — Best-performing brands (for contrast)

```
select brand_name,
       round(avg(rating)::numeric, 2) as avg_rating,
       count(*) as product_count,
       round(avg(price_usd)::numeric, 2) as avg_price
from products
group by brand_name
having count(*) >= 15
order by avg_rating desc
limit 10;
```

Brand	Avg Rating	Products	Avg Price
BondiBoost	4.78	34	28.62
StriVectin	4.66	24	69.75
FaceGym	4.65	16	83.38
Curlsmith	4.57	42	35.50
Bon Parfumeur	4.55	17	68.94
Sulwhasoo	4.54	23	85.26
iNNBEAUTY PROJECT	4.53	15	29.60
Dermalogica	4.52	51	60.29
Viktor&Rolf	4.51	19	108.00
Armani Beauty	4.50	54	80.59

Finding: Hair care (BondiBoost, Curlsmith) and skincare specialists (StriVectin, Dermalogica, Sulwhasoo) lead the top ranks. Combined with Q4, the pattern is consistent across three separate queries:

hair/skincare-focused brands perform well, fragrance brands underperform. That consistency is what makes this a real finding, not a coincidence in one chart.

Q6 — Does rating differ by skin type?

```
select skin_type,
       round(avg(rating)::numeric, 2) as avg_rating,
       count(*) as review_count
from reviews
where skin_type != 'Unknown'
group by skin_type
order by review_count desc;
```

Skin Type	Avg Rating	Review Count
Combination	4.31	291,804
Dry	4.29	99,379
Normal	4.29	69,286
Oily	4.29	66,005

Finding: Barely any gap here — 4.29 to 4.31 across every skin type. I'm not going to force a segmentation story that isn't in the data. This tells me satisfaction is driven by the product/category itself, not by who's using it — which actually supports focusing on Q3/Q4's category-level findings rather than a skin-type targeting strategy.

Q7 — Rating trend over time

```
select extract(year from submission_time::date) as year,
       round(avg(rating)::numeric, 2) as avg_rating,
       count(*) as review_count
from reviews
group by year
order by year;
```

Year	Avg Rating	Review Count
2008	4.46	1,931
2009	4.50	6,925
2010	4.53	9,689
2011	4.51	9,259
2012	4.51	8,845
2013	4.46	11,656
2014	4.37	12,578

Year	Avg Rating	Review Count
2015	4.33	14,081
2016	4.37	22,051
2017	4.40	38,617
2018	4.38	67,029
2019	4.27	91,819
2020	4.22	125,824
2021	4.31	91,712
2022	4.28	74,145
2023	4.21	14,970

Finding: Rating peaked in 2010 at 4.53 and has drifted down to the low 4.2s by 2023 — even as review volume grew more than 10x (9.7K reviews in 2010 vs 125K in 2020). I'm not claiming a single cause here, but two honest hypotheses: (1) more reviewers at scale naturally skews more critical/diverse, or (2) product proliferation diluted quality control. Worth flagging for further investigation — specifically whether this decline is broad-based or concentrated in Makeup/Fragrance, which we already flagged as weaker categories.

Summary — What I'd Tell the Business

Three things came out of this analysis that I'd take into a real meeting:

Finding: Fragrance and high-end skincare devices carry disproportionate rating risk — KILIAN Paris and NuFACE specifically stand out with real revenue exposure behind the low ratings, not just noise.

Finding: Overall rating has quietly declined over 13 years (4.53 → 4.21) even as the platform scaled massively — this deserves a deeper root-cause look, especially tied to the Makeup category dip.

Finding: Skin type isn't a meaningful lever for this problem — the fix is product/category-level, not audience-segment-level, which should shape where budget and attention actually go.

Next Steps

From here: sentiment analysis on review_text in Python (VADER) to see if written sentiment backs up the rating-based findings above, then building the 4-page Power BI dashboard to make all of this visual and explorable for a non-technical stakeholder.